

FULL STACK ASSIGNMENT SUBMISSION

Smart Leads Dashboard

MERN Stack Lead Management Platform

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Repository	smart-leads-dashboard

Submitted for the Full Stack Developer Internship opportunity.

1. Executive Summary & Objective

The **Smart Leads Dashboard** is a high-performance lead management solution built using a typesafe **MERN** stack architecture (React, Node.js, Express, MongoDB) with full **TypeScript** compliance. Designed to solve real-world sales operations challenges, it implements full CRUD operations, role-based security, multi-variable server-side filtering, and responsive containerization.

Assignment Evaluation Highlights

- **100% TypeScript Coverage:** Standardized data contracts, strictly avoiding `any`, using explicit compiler safety.
- **Advanced Server-Side Filtering:** Highly reactive, compound queries for Status, Source, and Search combined.
- **Strict Role-Based Access Control (RBAC):** Admin-only deletions and data mutations separate from standard Sales workflows.
- **Production Ready:** Multi-stage Docker environments and automated continuous deployment.

Key Live Project Resources

GitHub Repository

github.com/Jaswanth-Reddy-2006/smart-leads-dash-board

Production Frontend (Netlify)

smart-leads-dashboard-jaswanth.netlify.app

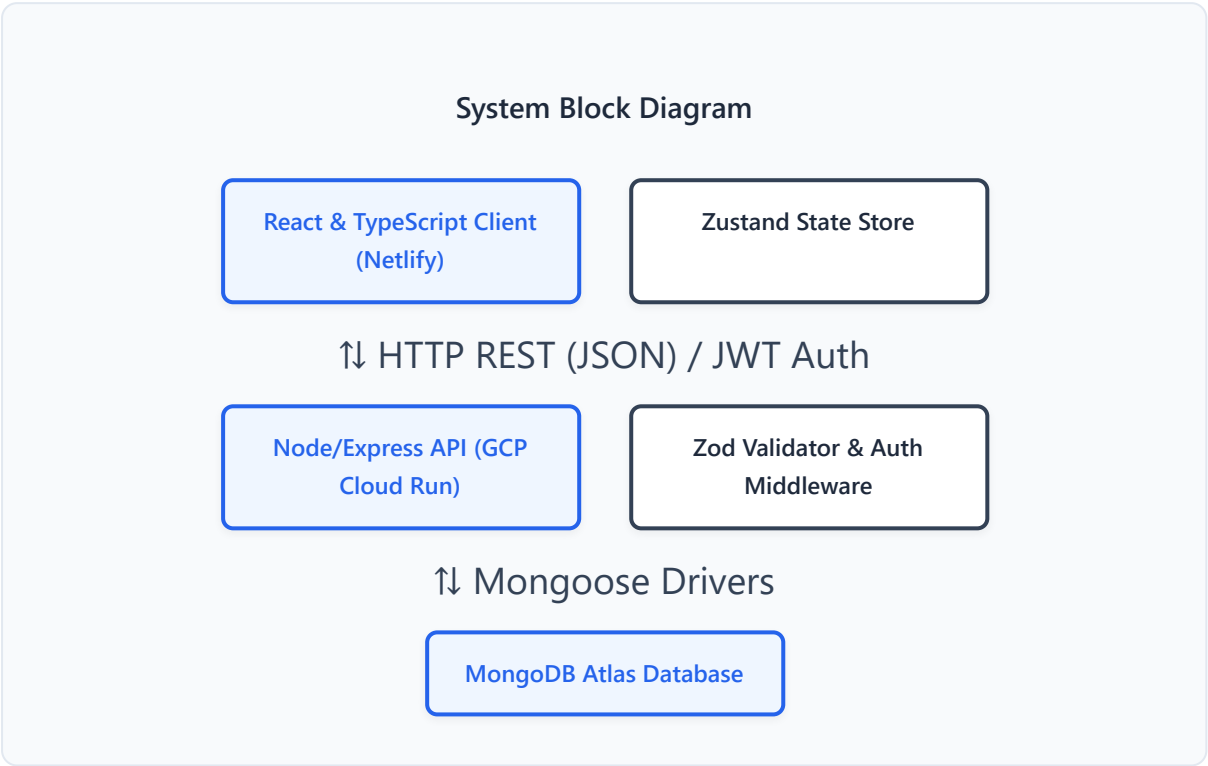
Production Backend (Cloud Run)

smart-leads-dashboard-api-qiwnf6awba-uc.a.run.app

API Health Endpoint

[/health](#) (Returns status: OK)

2. System Architecture & Tech Stack



Technology Stack Breakdown

Layer	Technology Chosen	Key Purpose
Frontend	React.js (v18), TypeScript	Strictly typed functional components & component lifecycle.
Styling	Tailwind CSS	Premium UI layout, native dark mode support, fluid interactions.
State Mgmt	Zustand	Minimalist, lightweight global store with JWT token hydration.
Backend API	Node.js, Express.js, TypeScript	Centralized routing, controller layer, validation, and error interceptors.
Validation	Zod Schema Validation	Enforces API payload integrity at runtime, mapping directly to TypeScript types.

Layer	Technology Chosen	Key Purpose
Database	MongoDB + Mongoose ODM	Flexible schema design, compound indexing, page aggregation queries.
Container	Docker & Docker Compose	Ensures uniform execution across dev and production.

Security Architecture

- **Password Hashing:** Users' password entries are salted and hashed using bcrypt (10 rounds) before DB persistence.
- **Stateless JWT Validation:** API server validates requests using signature matching for **Bearer** tokens in the request Authorization headers.
- **CORS Policy:** Enforced backend restrictions allow transactions only from registered UI origins.

3. API Specifications & Endpoints

The backend implements a standard RESTful API. Every request requires validation via Zod schemas, returning descriptive error status codes (`400` , `401` , `403` , `404` , `500`) with formatted error messages.

Route Map & Access Control

Method	Endpoint	Auth	Min Role	Functionality
POST	<code>`/api/auth/register`</code>	No	Any	Creates a new user profile with selected roles.
POST	<code>`/api/auth/login`</code>	No	Any	Issues a JWT token valid for 7 days.
GET	<code>`/api/auth/me`</code>	Yes	Any	Retrieves the current user's role and details.
GET	<code>`/api/leads`</code>	Yes	Sales	Lists leads with compound filters, search, and page values.
GET	<code>`/api/leads/:id`</code>	Yes	Sales	Retrieves a single lead's full detail.
POST	<code>`/api/leads`</code>	Yes	Sales	Creates a new sales lead. Validates fields with Zod.
PUT	<code>`/api/leads/:id`</code>	Yes	Sales	Updates an existing lead.
DELETE	<code>`/api/leads/:id`</code>	Yes	Admin	Deletes a lead database entry permanently.
GET	<code>`/api/leads/export`</code>	Yes	Sales	Retrieves filtered list in CSV download format.

Key API Payload Interfaces

```
// User Registration
interface IRegisterInput {
  name: string;
  email: string;
  password: string; // Min 6 chars
  role: 'admin' | 'sales';
}

// Pagination Metadata Envelope
interface IPaginatedResponse<T> {
  data: T[];
  pagination: {
    page: number;
    limit: number;
    total: number;
    totalPages: number;
    hasNextPage: boolean;
    hasPrevPage: boolean;
  }
}
```

4. Advanced Features & Configuration

Compound Filters & Debounced Search

The backend queries are built dynamically using MongoDB regex and filters. On the frontend, a custom debounce mechanism ensures that requests to search leads by name or email are delayed by 300ms. This prevents server overloading and provides a fluid user experience.

```
// Backend Controller Query Assembly Example
const query: any = {};
if (search) {
  query.$or = [
    { name: { $regex: search, $options: 'i' } },
    { email: { $regex: search, $options: 'i' } }
  ];
}
if (status) query.status = status;
if (source) query.source = source;
```

Environment Variables Configuration

Backend (`server/.env`)

```
NODE_ENV=production
PORT=5000
MONGO_URI=mongodb+srv://<username>:<password>@cluster0.gwdrfxt.mongodb.net/leads_db
JWT_SECRET=your_super_secret_jwt_key_change_this
JWT_EXPIRES_IN=7d
CORS_ORIGIN=https://smart-leads-dashboard-jaswanth.netlify.app
```

Frontend (`client/.env`)

```
VITE_API_BASE_URL=https://smart-leads-dashboard-api-qiwnf6awba-uc.a.run.app/api
```

Multi-Container Docker Configuration

A production-ready `docker-compose.yml` file orchestrates the local dev/prod stack, initializing isolated frontend, backend, and database environments with one terminal execution.

```
# Execution command:
docker compose up --build
```

5. Candidate Resume & Achievements

R Jaswanth Reddy

Computer Science Undergraduate | B.Tech CSE (GPA: 8.76) | Grad: 2028

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Technical Core Skills

- **Languages:** JavaScript (ES6+), TypeScript, Node.js, Python, C++
- **Frontend:** React.js, Tailwind CSS, HTML5, CSS3, Zustand, Context API
- **Backend & DB:** Express.js, MongoDB, Mongoose, REST APIs, Postgres, Firebase
- **Cloud & CI/CD:** Netlify, Vercel, Render, Google Cloud Platform (Cloud Run, Container Registry)
- **DevOps/Workflow:** Docker, Git, GitHub Actions, Linux Shell, VS Code/Cursor

Professional Experience & Internships

AI Frontend Developer Intern | EnviGuide (Jan 2026 – Mar 2026)

- Architected AI-driven dashboard user interfaces using React.js.
- Built responsive layout libraries and integrated complex AI model response APIs.
- Optimized rendering paths to scale dashboard capabilities under heavy data state mutations.

Achievements & Hackathons

- **Top 10 Finalist (Out of 1000+ Competitors):** Vibeathon Bangalore 2025. A 36-hour hackathon, creating an upskilling visualization platform for students.
- **Open Source Contributor:** Created *AlgoScope*—a premium interactive algorithm visualizer built with Canvas APIs.